Report on Cloud Architecture RePlate: Reinventing what goes on the plate

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https://github.com/abhayzap/Data-Science

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1. INTRODUCTION:

Food waste is a global issue, with 1.3 billion tons wasted each year. To address this dilemma, RePlate connects farmers, sellers, and customers using cloud technology, artificial intelligence, and real-time data. This article delves into RePlate's innovative architecture, which aims to streamline supply chains, eliminate waste, and build a sustainable food ecosystem via scalable cloud solutions.

2. Vision & Mission: A Smarter Food System:

Vision:

"RePlate harnesses cloud technology to combat food waste sustainably, ensuring scalability, real-time insights, and seamless collaboration across the food supply chain."

Mission:

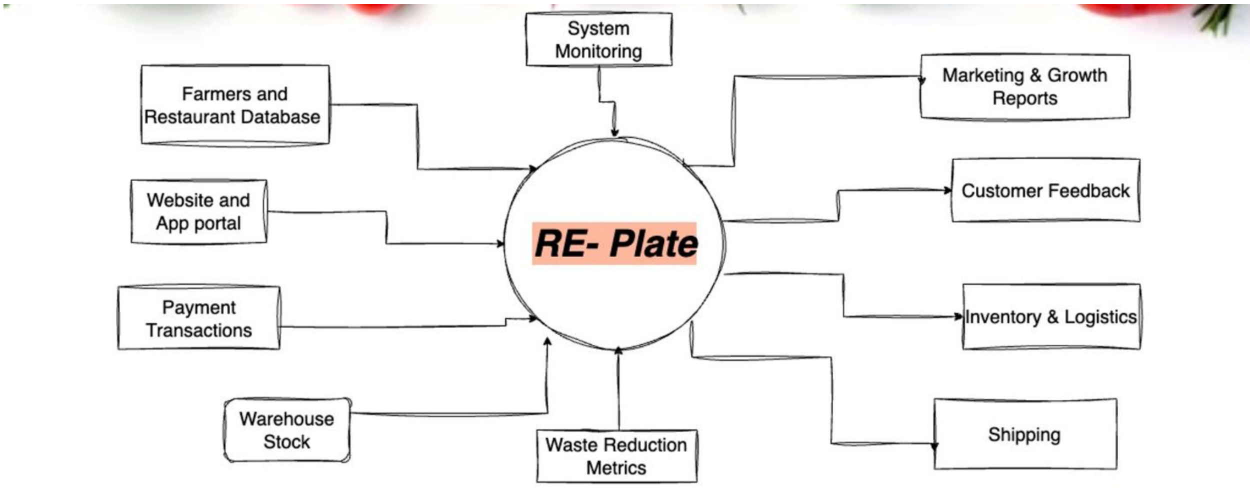
RePlate integrates Azure cloud services, AI, and IoT to:

- Optimize supply chains via predictive analytics and real-time tracking.
- Connect stakeholders through a scalable marketplace.
- Reduce operational costs and expand access to rescued produce.

3. Objectives:

- ➢ **Supply Chain Optimization**: Minimize waste using IoT sensors and AI-driven forecasting.
- ➢ **Unified Marketplace**: Enable real-time collaboration between farmers, vendors, and consumers.
- ➢ **Automation**: Streamline workflows with serverless computing (Azure Functions).
- ➢ **Compliance & Security**: Ensure data protection and adherence to food safety regulations.

4. Data Sources:



1. Farmers and Restaurant Database:

Purpose:
Monitors suppliers (farmers, restaurants) and their surplus produce availability.
Historical sourcing data is recorded to discover patterns (such as seasonal surplus).

Data type:
Structured data (SQL tables) include supplier profiles, product categories, quantities, and pickup schedules.

Use Case:
Predict surplus produce availability with AI-powered forecasting.
Match surplus inventory with vendors or organizations in real time.

2. Website and App Portal:

Purpose:
Record user interactions, orders, and navigation behaviors.
Provides seamless connectivity between customers, vendors, and suppliers.

Data Type:
Semi-structured logs (JSON/CSV) for clickstream data, session durations, and cart abandonment rates.
Transactional data includes orders, subscriptions, and user preferences.

Use case:
Analyze consumer demand trends to optimize inventory distribution.
Create personalized recommendations (e.g., "rescued produce" alerts).

3. Payment Transactions:

Purpose:
Logs financial interactions, including purchases, refunds, and invoices.

Data Type:
Structured data (SQL tables): Transaction IDs, payment methods, timestamps.

Use Case:
Track revenue streams and identify high-demand products.
Ensure compliance with financial regulations (e.g., tax reporting).

4. Warehouse Stock (IoT Sensors):

Purpose:
Monitors real-time inventory levels, storage conditions (temperature, humidity), and spoilage risks.

Data Type:
Time-series data (IoT sensors): Stock quantities, expiration dates, environmental metrics.
Event-triggered alerts (e.g., temperature spikes).

Use Case:
Trigger automated alerts for expiring stock to prioritize redistribution.
Optimize warehouse layouts based on perishability trends.

5. Marketing & Growth Reports:

Purpose:
Measures campaign effectiveness and customer acquisition metrics.

Data Type:
Structured analytics (Power BI): ROI dashboards, conversion rates, customer demographics.

Use Case:
Refine outreach strategies (e.g., targeting regions with high food waste).
Track growth in rescued produce distribution.

6. Customer Feedback:

Purpose:
Gathers reviews, ratings, and complaints to improve service quality.

Data Type:
Unstructured text (NLP-processed): Sentiment analysis, keyword trends.
Structured ratings (1–5 stars).

Use Case:
Identify pain points in the supply chain (e.g., delivery delays).
Enhance user experience through actionable feedback.

7. Logistics and Shipping Systems:

Purpose:
Tracks delivery routes, carrier performance, and SLA compliance
.
Data Type:
Geospatial data (GPS): Delivery vehicle locations, route efficiency.
Structured logs: ETAs, dispatch times, delivery confirmations.

Use Case:
Optimize delivery routes using predictive algorithms.
Reduce carbon footprint by minimizing fuel consumption.

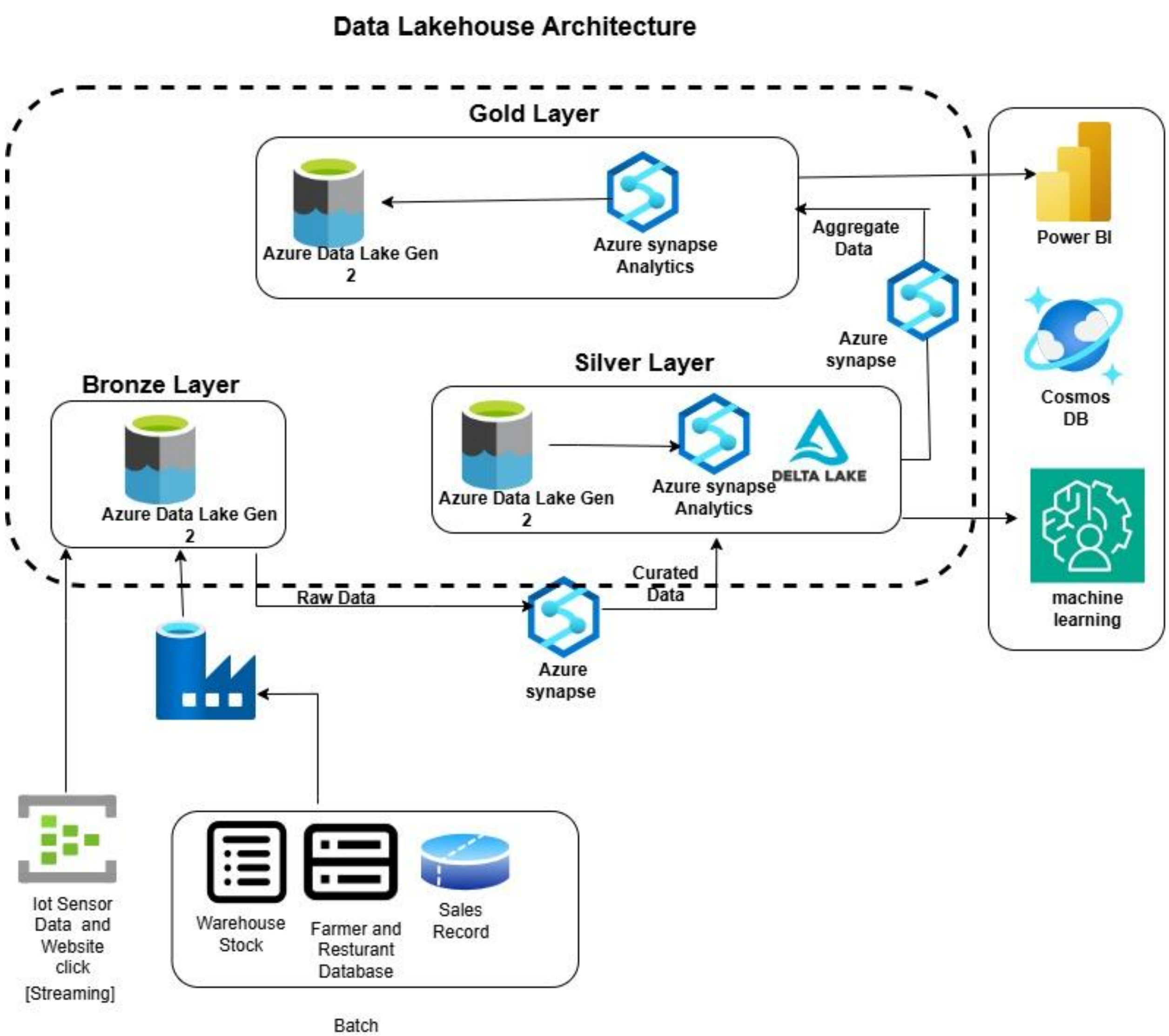
8. External APIs:

Purpose:
Integrates external data like weather forecasts to predict farming yields or delivery disruptions.

Data Type:
Semi-structured (JSON/XML): Temperature, rainfall, storm alerts.

Use Case:
Adjust surplus predictions during adverse weather.
Proactively reroute deliveries to avoid delays.

5. LAKEHOUSE ARCHITECTURE:



- **Data Sources and Ingestion:**
  The system uses Azure Data Factory to ingest both streaming data (from IoT sensors and website clicks) and batch data (from warehouse inventories, restaurant databases, and sales records). This ensures large-scale capture of both real-time and historical data.

- **Storage Layers:**
  Data is stored in Azure Data Lake Gen2, using a multi-layered technique.

  The Bronze Layer stores raw, unprocessed data.

Silver Layer: This layer contains data that has been checked and cleaned using Delta Lake for ACID transactions.

The Gold Layer stores aggregated, analytics-ready data for business insights.

- **Data Processing:**
  Azure Synapse and Delta Lake handle data transformation using ETL pipelines. Data progresses from bronze to silver to gold, becoming getting refined and enriched.

- **Analytics:**
  Using Azure Synapse Analytics, enterprises can run scalable, interactive searches & advanced analytics on curated datasets, enabling business information and reporting needs.

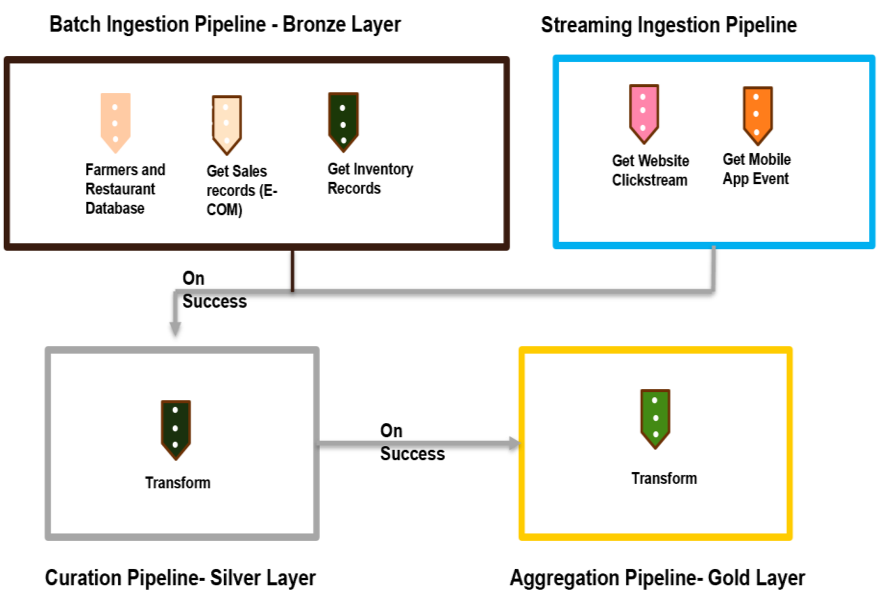
- **Data Consumption and Usage:**
  The revised data is then utilized by:

  Power BI is used for dashboarding and reporting.

  Cosmos database for operational analytics and applications.

  Machine learning models for predictive analytics and AI-powered decisions.

6. Pipeline Design:



1. **Batch Ingestion:** Pipeline sources include farmers' databases, e-commerce sales, and warehouse inventory.

Workflow:
Extractions were scheduled using Azure Data Factory.
Raw data is stored in the Bronze Layer.

2. Streaming Ingestion Pipeline: Sources include website clicks, mobile app events, and IoT sensors.

Workflow:
Azure Event Hub enables real-time data ingestion.
Stream Analytics is used to discover anomalies.

3. Curation Pipeline (Silver Layer):
Tasks include schema normalization, data enrichment, and deduplication.
Output: Delta Lake tables have been curated for downstream analytics.

4. Aggregation Pipeline (Gold Layer) Tasks:
Create waste-reduction metrics (such as "rescued produce per region").
Use machine learning models to analyze client turnover.
Output includes Power BI dashboards and Synapse SQL Pool tables.

7. Failure Handling: Ensured Resilience:

Common problems include data corruption, network failures, and misconfigured pipelines.

Mitigation:
- Automated retry with exponential backoff.
- Azure Monitor generates alerts for operational teams.
- Roll back to the last successful checkpoint.

8. Conclusion:

RePlate's cloud-based data architecture seamlessly connects IoT, transactional systems, and artificial intelligence to minimize food waste. By converting raw data into real-time insights, it improves supply chains, minimizes spoilage, and enables stakeholders to act quickly. This scalable, Azure-powered architecture not only promotes sustainability, but also lays the groundwork for technology-driven solutions in global food systems.